

Gröbner Bases Native to 'pseudo'-Hodge Algebras, with an Application to Bideterminants

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► Equivalently: Does f belong to the ideal generated by $h_1 \dots h_k$

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- ▶ UNDERSTATEMENT: Gröbner bases are very useful

Computational Complexity Theory (CCT)

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- ▶ Algebraic version of P vs NP is called permanent vs determinant

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- ▶ Conjecture: $\text{dc}(\text{perm}_n) = n^{\omega(1)}$

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- ▶ Sometimes you get intuition by computing for small $n = 2, 3, \dots$
- ▶ **Gröbner bases are well-suited to all**: they give theoretical insight as well as are the key tool in effective methods

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- ▶ Gröbner bases tend to **obscure symmetry**!

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- ▶ One can obtain lower bounds on **decision tree complexity** by obtaining Betti numbers of semi-algebraic sets
- ▶ Requires computing Gröbner bases of D-ideals (ideals in the Weyl algebra):

$$W_n := \mathbb{C} \left[\left\{ X_i, \frac{\partial}{\partial X_i} \right\}_{i \in [n]} \right] / \left\langle \left\{ \frac{\partial}{\partial X_i} \cdot X_i - X_i \cdot \frac{\partial}{\partial X_i} - 1 \right\}_{i \in [n]} \right\rangle$$

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Question

Develop a Gröbner basis theory which takes advantage if variety corresponding to your ideal is 'determinantal'.

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 - ▶ **straightening**: the product of two standard monomials can be expressed as a linear combination of standard monomials
- ▶ A is an **ASL** if the straightening has some additional structure
- ▶ ASLs arise as coordinate rings of algebraic varieties, e.g. Grassmannians, determinantal varieties, flag varieties, Schubert varieties

Bideterminants (products of minors)

Denoting

$$(r_1, \dots, r_k \mid c_1, \dots, c_k) := \det \left(\begin{bmatrix} x_{r_1, c_1} & \cdots & x_{r_1, c_k} \\ \vdots & \ddots & \vdots \\ x_{r_k, c_1} & \cdots & x_{r_k, c_k} \end{bmatrix} \right),$$

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The combinatorial datum below (**bitableau**)

$$\left[\begin{array}{|c|c|} \hline r_1^{(1)} & r_1^{(2)} \\ \hline \vdots & \vdots \\ \hline \vdots & \vdots \\ \hline \vdots & r_{\lambda_2}^{(2)} \\ \hline r_{\lambda_1}^{(1)} & \\ \hline \end{array} \cdots \begin{array}{|c|} \hline r_1^{(p)} \\ \hline \vdots \\ \hline r_{\lambda_p}^{(p)} \\ \hline \end{array} \mid \begin{array}{|c|c|} \hline c_1^{(1)} & c_1^{(2)} \\ \hline \vdots & \vdots \\ \hline \vdots & \vdots \\ \hline \vdots & c_{\lambda_2}^{(2)} \\ \hline c_{\lambda_1}^{(1)} & \\ \hline \end{array} \cdots \begin{array}{|c|} \hline c_1^{(p)} \\ \hline \vdots \\ \hline c_{\lambda_p}^{(p)} \\ \hline \end{array} \right],$$

defines the following product of minors (**bideterminant**)

$$\prod_{i=1}^p (r_1^{(i)}, \dots, r_{\lambda_i}^{(i)} \mid c_1^{(i)}, \dots, c_{\lambda_i}^{(i)}).$$

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- ▶ Thus **Standard bideterminants form an $\mathbb{F}$ -linear basis** of $\mathbb{F} \left[\{X_{i,j}\}_{i \in [m], j \in [n]} \right]$

Bideterminants give Hodge Algebra

► Define

$$A = \mathbb{F}[X_{1,1}, X_{1,2}, X_{2,1}, X_{2,2}, Y] / \langle X_{1,2}X_{2,1} - X_{1,1}X_{2,2} + Y \rangle$$

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- As seen before, $A \cong \mathbb{F}[X_{1,1}, X_{1,2}, X_{2,1}, X_{2,2}]$

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- ▶ This generalizes
 - ▶ poly ring with one variable for each minor of $n \times m$ matrix
 - ▶ quotient by relations between minors
 - ▶ gives ASL structure to the co-ordinate ring of $n \times m$ matrices
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 - ▶ **quotient by relations** between minors
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 - ▶ **standard monomials** correspond to **standard bitableaux**
- ▶ Advantage - smaller expressions for 'determinant-like' polynomials; bideterminants reflect symmetries coming from the action (representation theory) of GL_n

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Question

Can we build a theory of Gröbner bases 'native' to p-ASLs, i.e. Gröbner theory without referencing the ideal J ?

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- ▶ Product of standard monomials not necessarily standard, might require **straightening**
 - ▶ How do you define **term order**?
 - ▶ How would you define **division** of monomials?
 - ▶ What plays the role of **monomial ideals**?

Term Order & Division

► A **p-ASL term order** on a p-ASL A is a total order $\prec$ on standard monomials in A such that

► $1 \preceq m$

► If $a \prec b$ and $c \preceq d$, and $ac, bd \neq 0$, then

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- ▶ When does standard monomial m **divide** m' :

- ▶ **ordinary division** in the polynomial ring, or

- ▶ m divides m' if **there exists** standard monomial f such that

$$\text{LM}(mf) = m'$$

Auxilliary Algebra of Leading Terms

- Given p-ASL A , **algebra of leading terms** w.r.t. A is another p-ASL A_{lt} on the same variables, and the same standard monomials such that for standard monomials m, m'

$$\pi_{lt}(m) \cdot \pi_{lt}(m') = \overbrace{0}^{\text{no straightening}} \quad \text{or} \quad \underbrace{\pi_{lt}(\text{LT}(mm'))}_{\text{leading term of straightening}}$$

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Proposition

Every p-ASL A admits two algebras of leading terms – A_{gen} where the product is never 0, and, A_{disc} where product is 0 unless mm' is also a standard monomial.

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 - ▶ $\langle \{\pi_{lt}(LM(g)) : g \in G\} \rangle = \langle \{\pi_{lt}(LM(f)) : f \in I\} \rangle$ (standard monomial ideals in A_{lt})

Our Main Result

Theorem (GN-Grochow and **Abhiram Natarajan** [2025])

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Reduced	✓	✓
Universal	✓	✓
Syzygies	✓	✓
Algorithms	✓	✓
Krull Dim.	✓	✓

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<i>Existence</i>	✓	✓
<i>Reduced</i>	✓	✓
<i>Universal</i>	✓	✓
<i>Syzygies</i>	✓	✓
<i>Algorithms</i>	✓	✓
<i>Krull Dim.</i>	✓	✓

Corollary (GN-Grochow and Abhiram Natarajan [2025])

The algebra of bideterminants has a p -ASL term order, thus we have a Gröbner basis theory (called bd -Gröbner bases).

Applications to Bideterminant Algebra

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Takeaway

1. *Given all our machinery, the proof is one-line*
2. *In the ordinary case, universal Gröbner basis are known only for maximal minors and minors of size 2*

Future Work

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- ▶ Investigate development of bipermanent Gröbner basis theory (codimension of singular locus of permanent hypersurface is unknown!)

References

- A. Björner and L. Lovász. Linear decision trees, subspace arrangements and Möbius functions. *J. Amer. Math. Soc.*, 7(3):677–706, 1994. ISSN 0894-0347,1088-6834. doi: 10.2307/2152788. URL <https://doi.org/10.2307/2152788>.
- C. de Concini, D. Eisenbud, and C. Procesi. Young diagrams and determinantal varieties. *Invent. Math.*, 56(2):129–165, 1980. ISSN 0020-9910,1432-1297. doi: 10.1007/BF01392548. URL <https://doi.org/10.1007/BF01392548>.
- C. De Concini, D. Eisenbud, and C. Procesi. *Hodge algebras*, volume 91 of *Astérisque*. Société Mathématique de France, Paris, 1982. With a French summary.
- J. Désarménien, J. P. S. Kung, and G.-C. Rota. Invariant theory, Young bitableaux, and combinatorics. *Advances in Math.*, 27(1):63–92, 1978. ISSN 0001-8708. doi: 10.1016/0001-8708(78)90077-4. URL [https://doi.org/10.1016/0001-8708\(78\)90077-4](https://doi.org/10.1016/0001-8708(78)90077-4).
- J. A. Grochow and Abhiram Natarajan. Gröbner bases in a bideterminant-like basis in the weyl algebra. In Preparation.
- J. A. Grochow and Abhiram Natarajan. Gröbner bases native to term-ordered commutative algebras, with application to the hodge algebra of minors, 2025. URL <https://arxiv.org/abs/2510.11212>. 107 pages, Accepted at MEGA, 2026. Submitted to Memoirs of AMS.
- E. W. Mayr and A. R. Meyer. The complexity of the word problems for commutative semigroups and polynomial ideals. *Advances in mathematics*, 46(3):305–329, 1982.
- T. Oaku. An algorithm of computing b-functions. *Duke Math. J.*, 87(1):115–132, 1997. ISSN 0012-7094,1547-7398. doi: 10.1215/S0012-7094-97-08705-6. URL <https://doi.org/10.1215/S0012-7094-97-08705-6>.
- A. Wigderson. *Mathematics and Computation : A Theory Revolutionizing Technology and Science*. Princeton University Press, New Jersey :, 2019. ISBN 9780691192543. URL <https://digital.casalini.it/9780691192543>.